

Oscillations and Waves.

XI

most mechanical devices that we use have oscillating parts - pistons. ~~As a water~~ Wave motion is an oscillatory phenomenon. A water wave travels across the sea, water molecules vibrate up and down. Sound waves travel through the air, air molecules vibrate back and forth. Em waves (radio, light, X-rays etc), the magnetic field oscillates as the wave moves forward. Materials particles exhibit wave phenomena, a fact that is fundamental in the quantum theory of matter.

Examples

- The motion of the moon about the Earth.
- Electrons in the atom about the nucleus.
- pendulum
- strings, bell, drum
- A.C. voltage.

A cycle is defined as motion that brings back the system back to its original state or position.

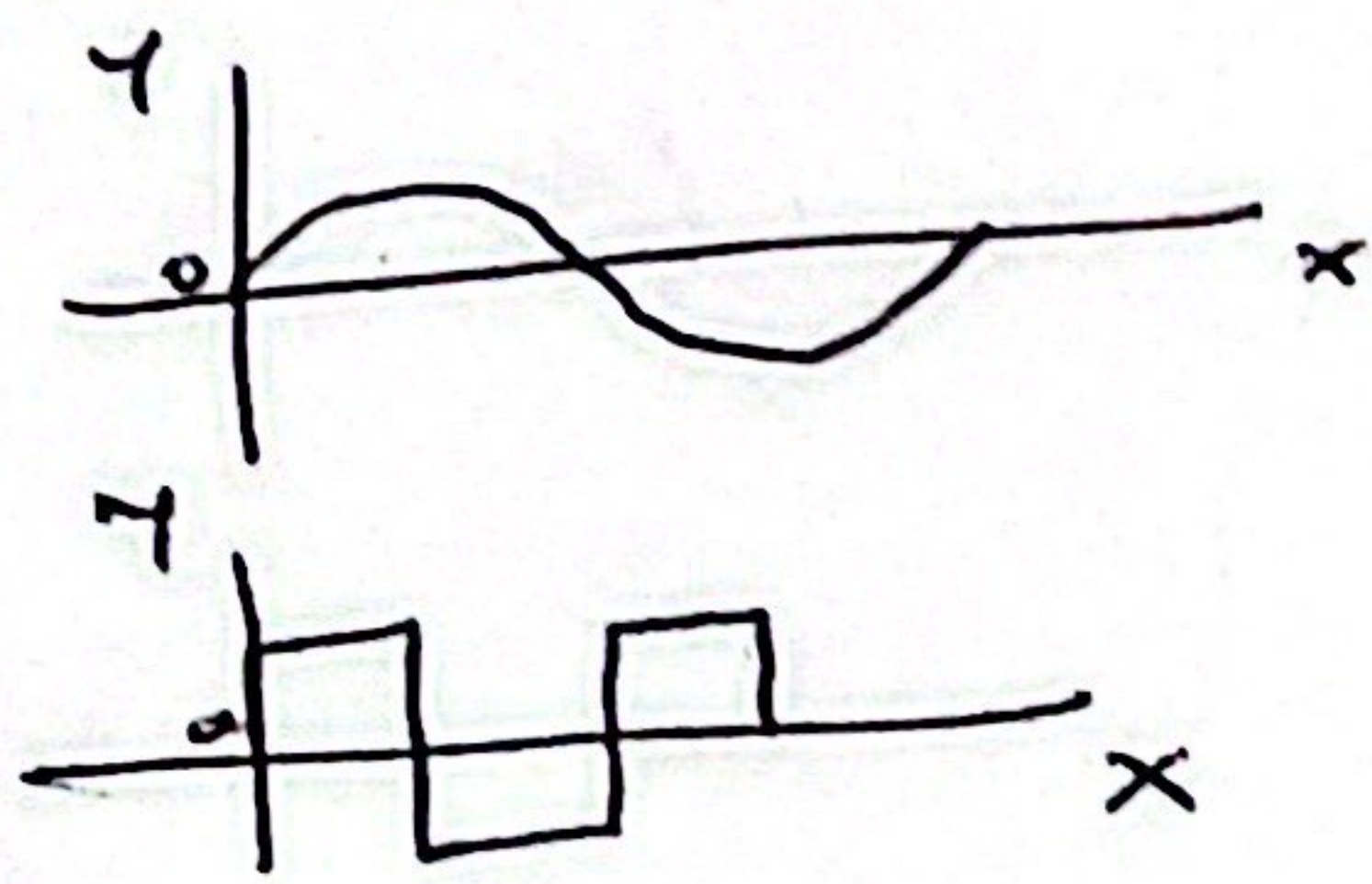
The length of time to execute a cycle is called the period T (s).

The no of cycles each second is frequency ν , related to the period by

$$\nu = \frac{1}{T}$$

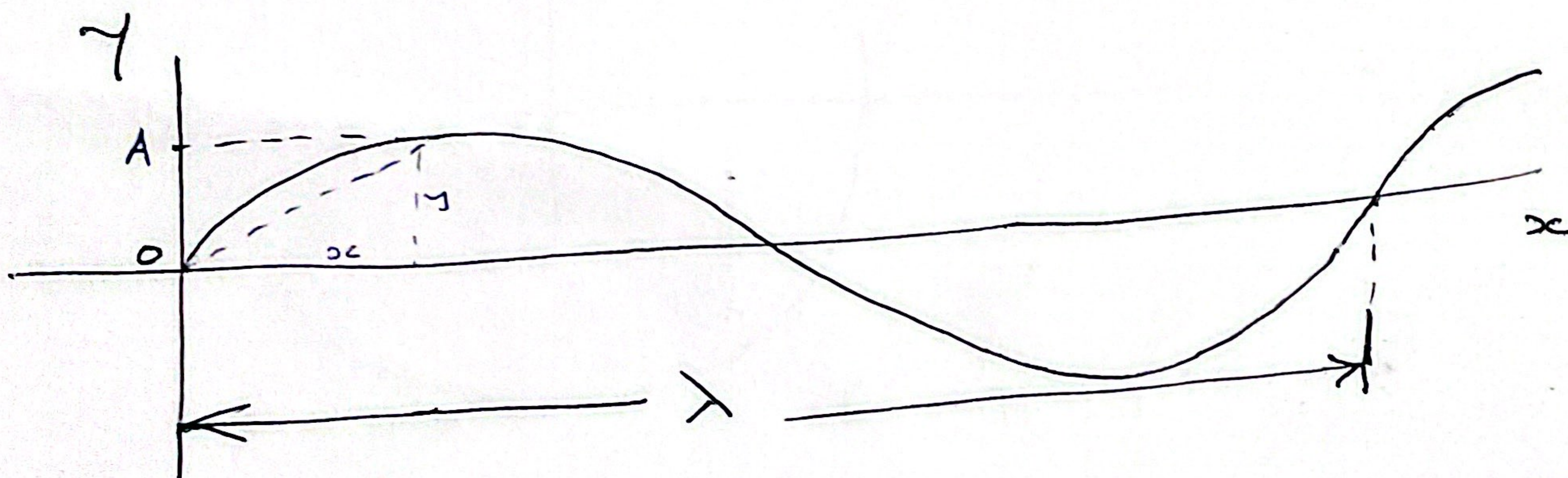
$$T = \frac{1}{\nu} \text{ Hz (1/s)}$$

Wave forms - Sine,
- Square



Wavelength λ , is the product of the speed of the wave V and the period T of the driving force of vibration.

$$\lambda = VT$$



Letting $T = \frac{1}{\nu}$, then $\lambda = \frac{V}{\nu}$ or

$$V = \lambda \nu$$

(relation of wave speed, wavelength and frequency).

note

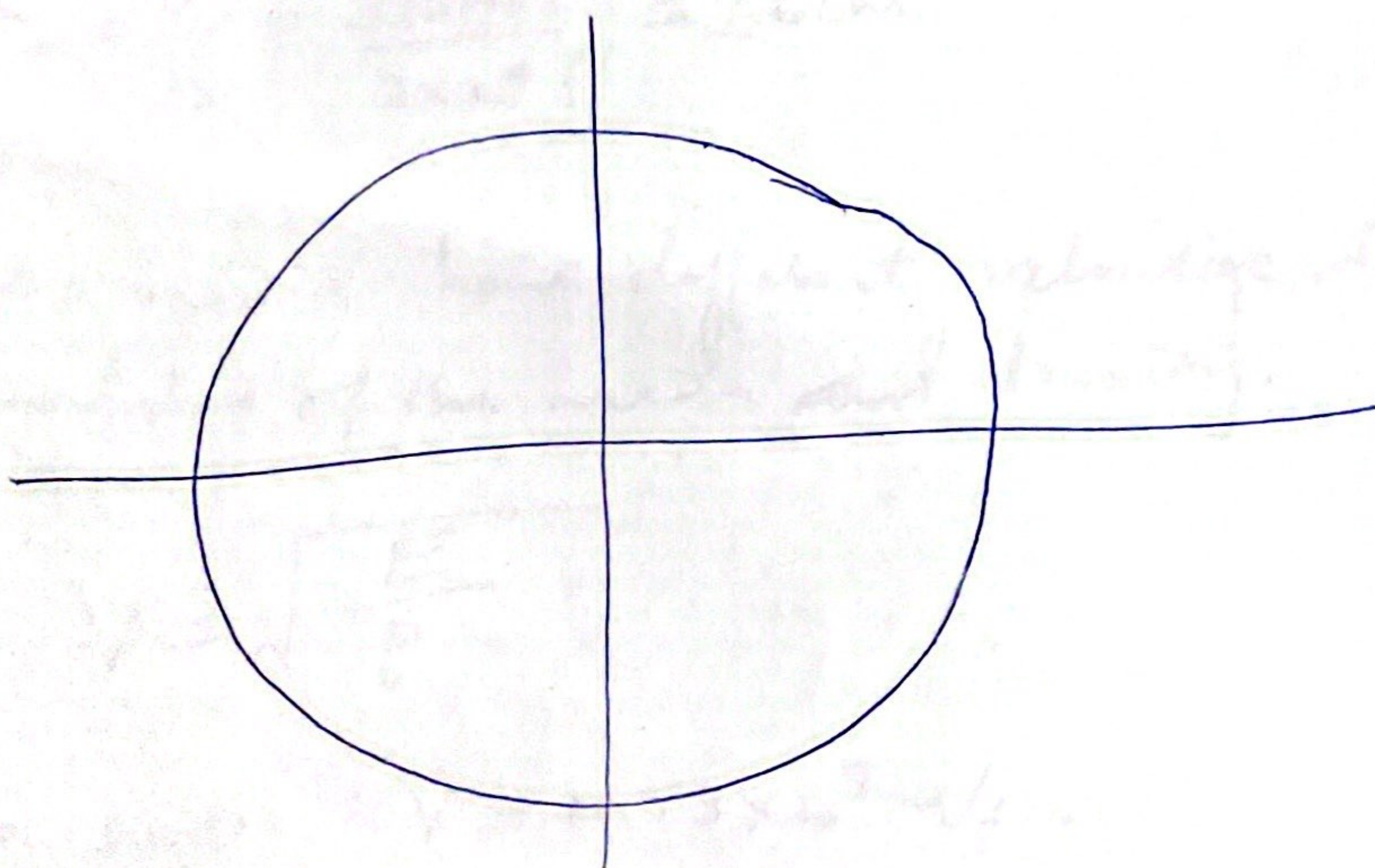
- ☐ ν or T depends on the source of vibrations
- ☐ V depends on the medium of propagation
- ☐ λ depends on V and ν , hence both medium and source.

Displacement $y = A \sin(\omega t - \phi) = A \sin 2\pi \left(\frac{x}{\lambda} - \frac{t}{T} \right)$

$$\omega = 2\pi\nu = \frac{2\pi}{T} \quad \text{and} \quad k = \frac{2\pi}{\lambda}$$

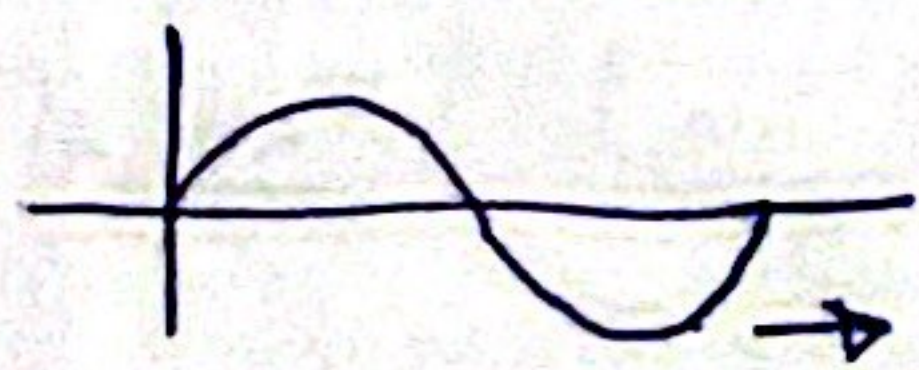
Gives a compact form of the displacement

$$y = A \sin(kx - \omega t)$$

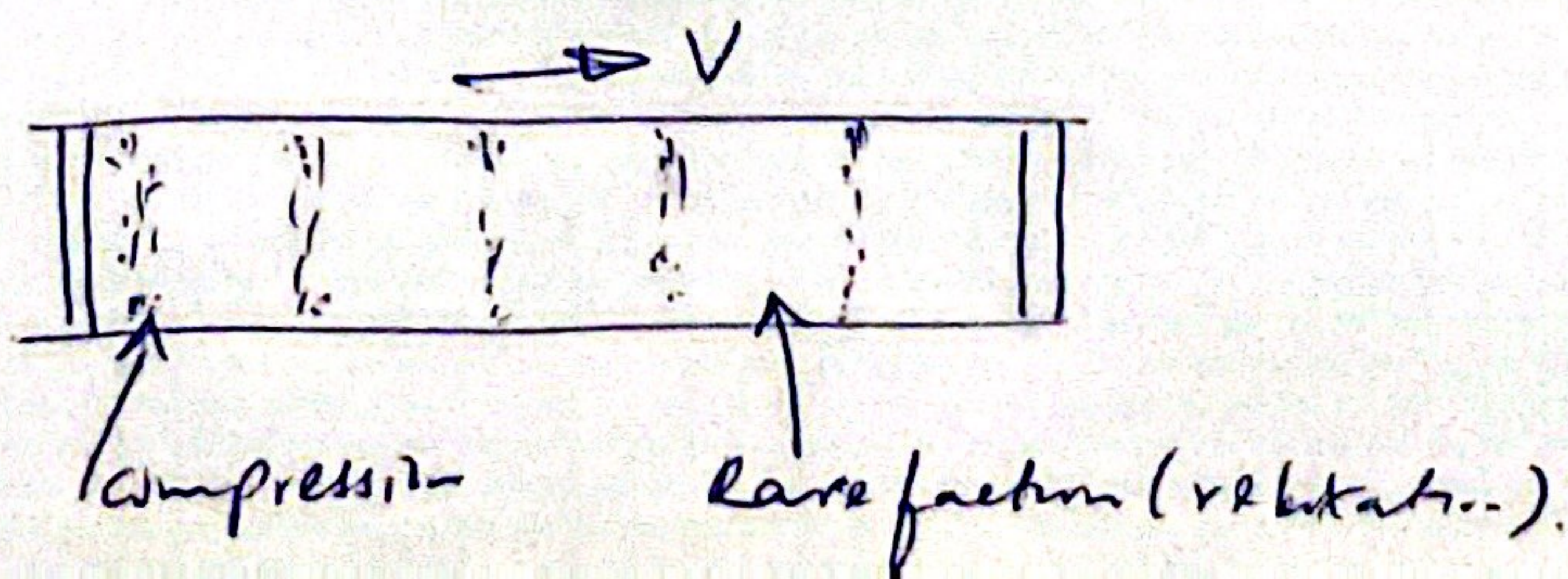


Types of waves

transverse — displacement of particles is $\perp$ to the direction of propagation. Eg. light (Em), water waves.
String waves



Longitudinal — displacement is parallel to the direction of propagation. Eg. spring compression, sound waves



Sound waves

In air $V_{\text{sound}} \approx 330 \text{ m/s}$ which is independent of frequency.

The pressure changes provoke a nervous response and we hear the sound in the range $16 \text{ Hz} - 20000 \text{ Hz}$. The wavelengths

are $\lambda_1 = \frac{V}{\nu_1} = \frac{330 \text{ m/s}}{16 \text{ s}^{-1}} \approx 20 \text{ m}$

$$\lambda_2 = \frac{V}{\nu_2} = \frac{330 \text{ m/s}}{2 \times 10^4 \text{ s}^{-1}} = 1.6 \text{ cm}.$$

Different media have different velocities depending on the young's modulus of the media and density.

$$V = \sqrt{\frac{E}{\rho}}$$

E.g. Sea water $V = 1.53 \times 10^3 \text{ m/s}$

Iron $V = 5.13 \times 10^3 \text{ m/s}$

Example

What is the wavelength of a 10 kHz sound wave in iron? What would be the wavelength of the same in air?

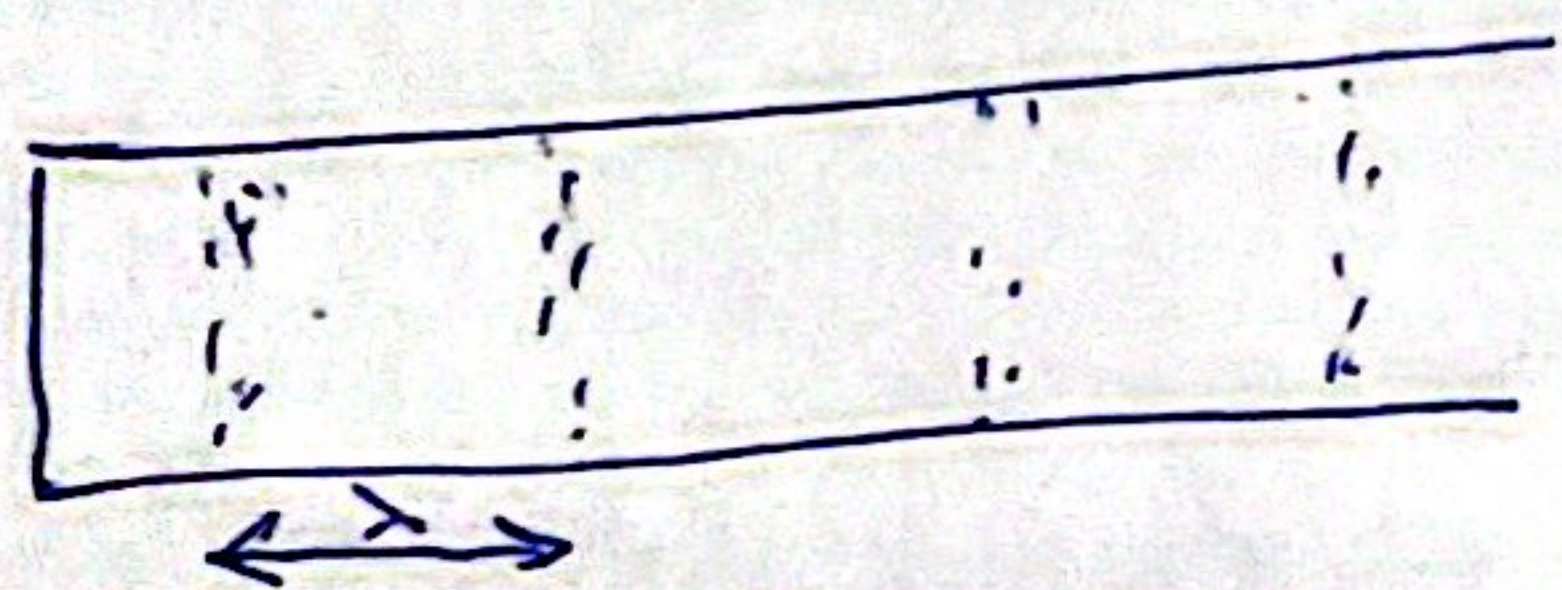
Kinetic theory predicts for the speed of sound in a gas as

$$V_{\text{sound}} = \sqrt{\frac{\gamma P}{\rho}} = \sqrt{\frac{\gamma K T}{m}} = \sqrt{\frac{\gamma}{3}} V_{\text{rms}}$$

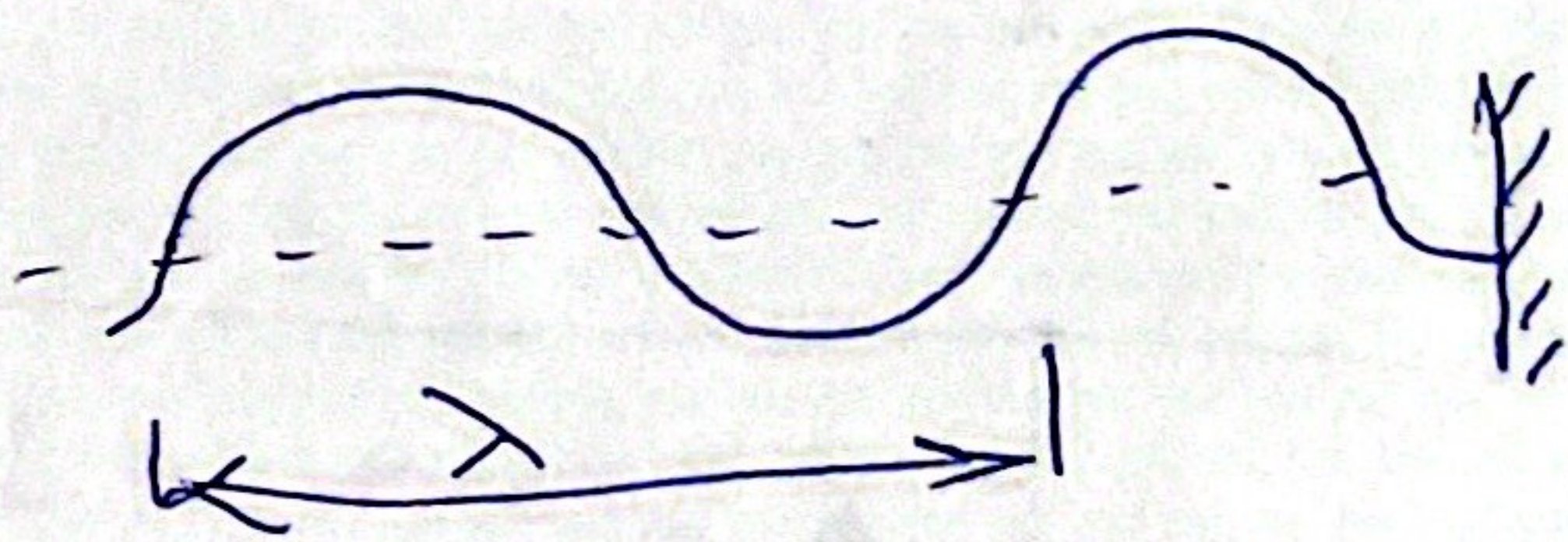
For diatomic molecules $\gamma = 7/5$ such as air so that

$$V_{\text{sound}} = 0.68 V_{\text{rms}}$$

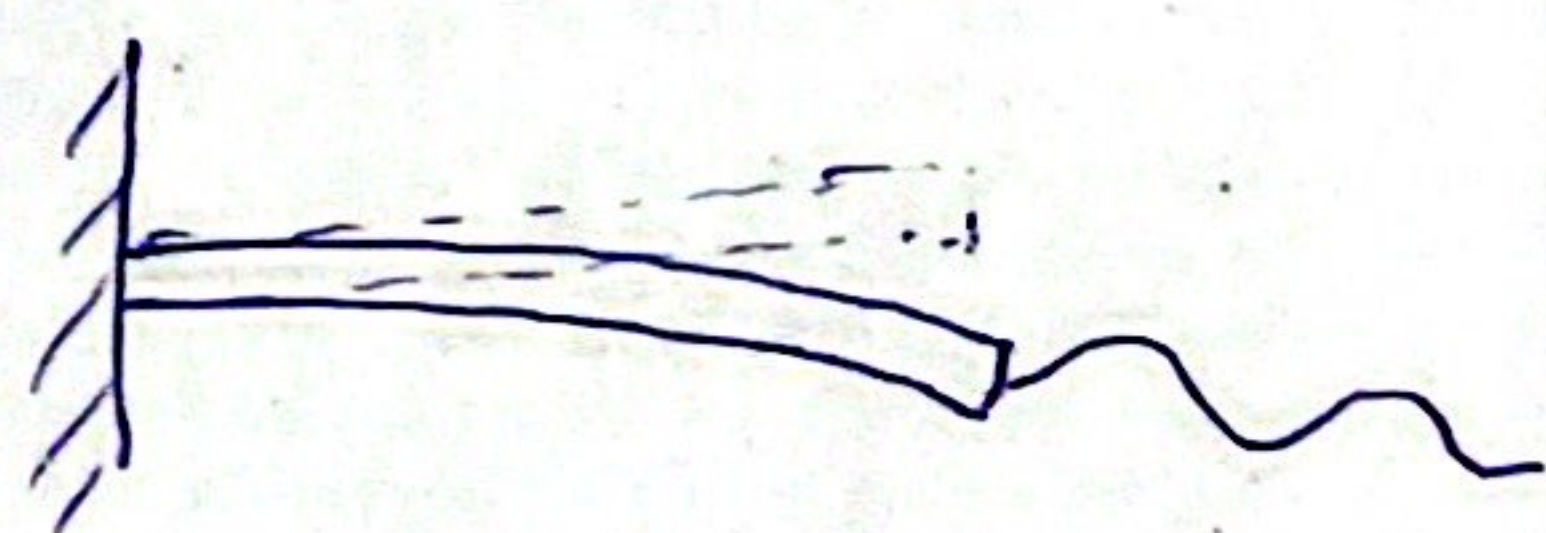
Vibration in pipes, strings, rods



Pipes



Strings

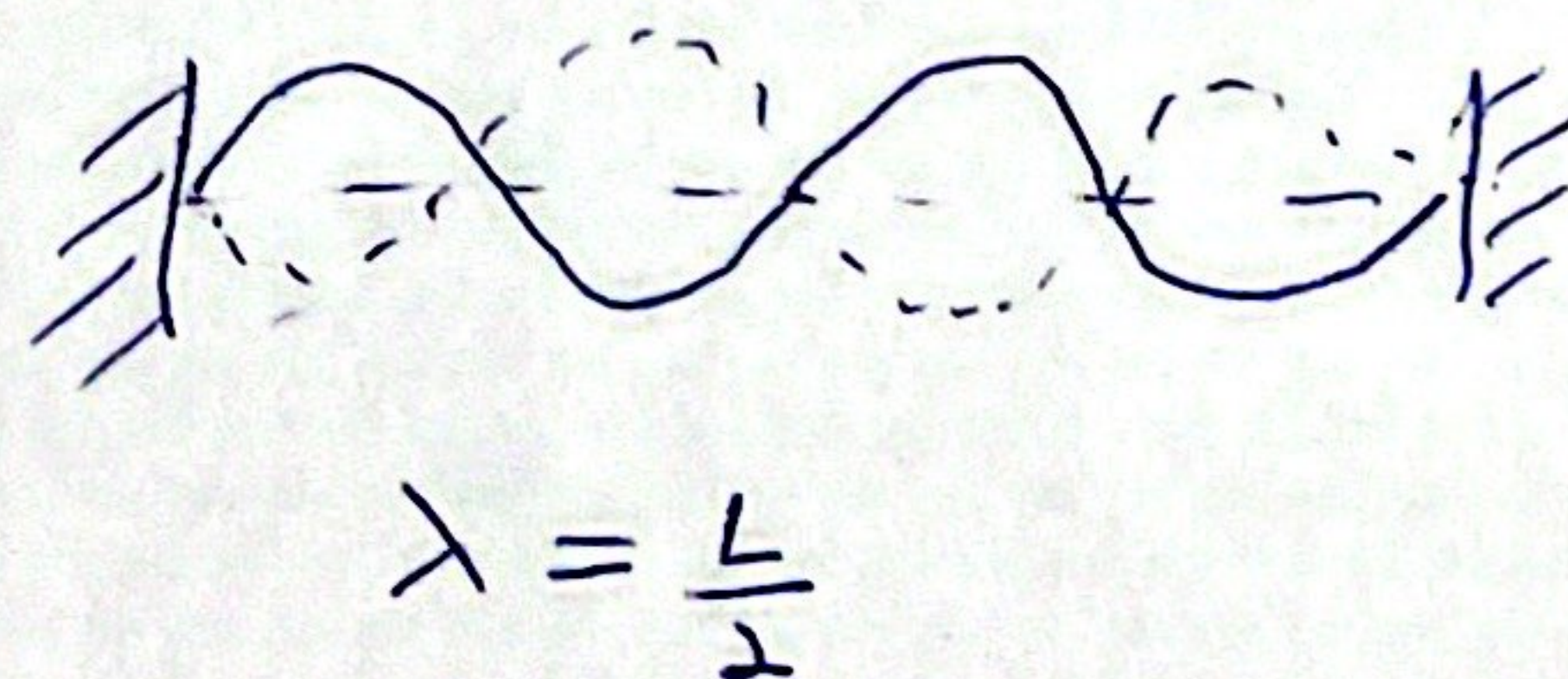
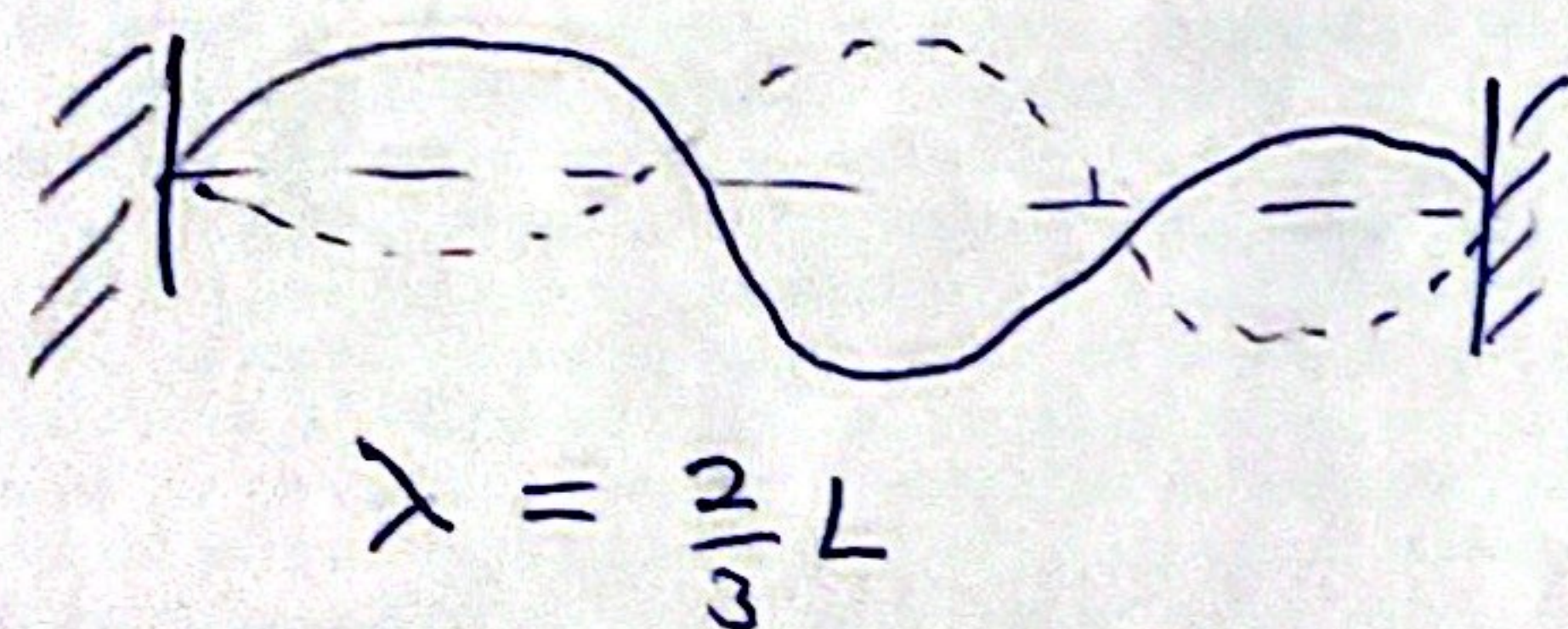
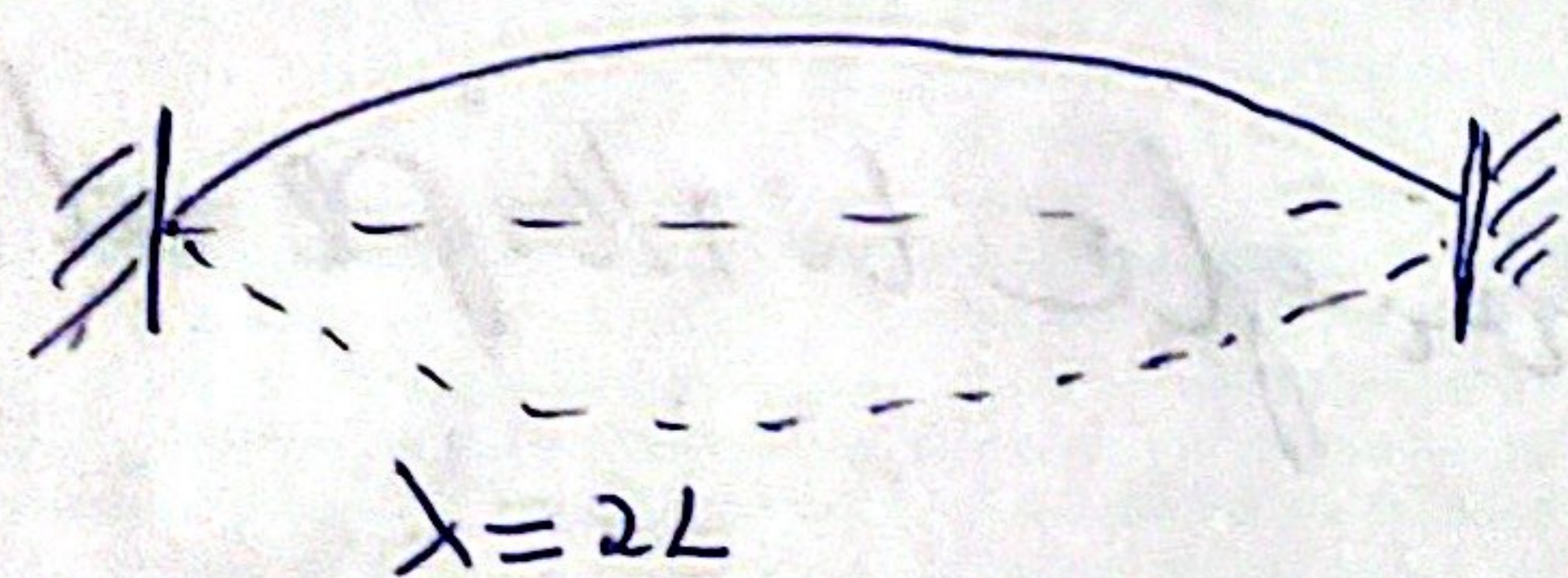
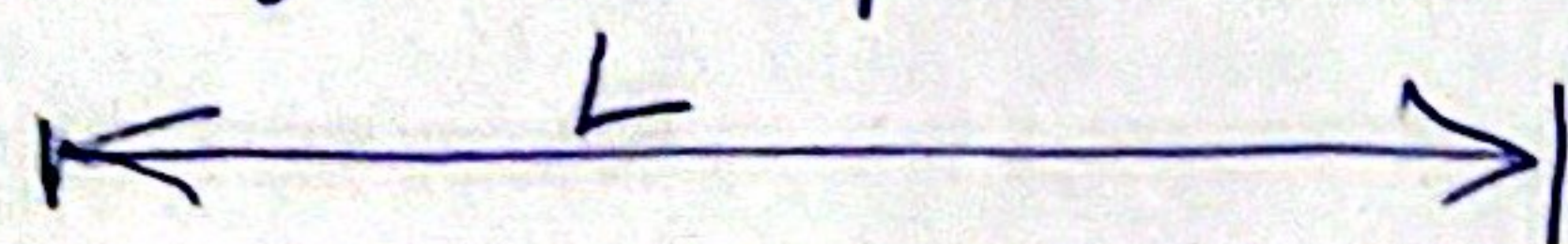


rods

$$V = \lambda \nu \text{ in all cases.}$$

Standing waves.

Time development of a disturbance between fixed supports or ends produces a standing wave. These waves occur for situations other than transverse vibrational waves such as in standing sound waves between reflecting walls or EM waves reflected from a pair of surfaces to form standing waves.



In general $\lambda = \frac{2}{n}L$, $n = 1, 2, 3, \dots$ for standing waves.

Example

What is the lowest frequency of the standing wave of sound that can be set up between walls separated by 25m?

Sol.

Lowest frequency is of $\lambda = 2L$. $\therefore \lambda = 2 \times 25\text{m} = 50\text{m}$.

$$\text{frequency } \nu = \frac{330\text{m/s}}{50\text{m}} = 6.6 \text{ Hz.}$$

General wave equation

1 dimensional $\left\{ \begin{array}{l} \frac{\partial^2 y}{\partial x^2} = \frac{P}{T} \frac{\partial^2 y}{\partial t^2} \quad \text{with } \frac{P}{T} = \frac{1}{v^2} \\ \text{or} \\ \frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2} \end{array} \right.$

3 dimensional ? $\rightarrow$ google wikipedia

- mass on a spring $T = 2\pi \sqrt{\frac{m}{k}}$

- circular motion

- oscillatory motion $T = 2\pi \sqrt{\frac{L}{g}}$

Resonance

This phenomenon occurs at a specific angular frequency ω and is characterized by a large amplitude of vibration. Standing waves in most cases represent resonance. A wire vibrating at a frequency and wavelength whose product yields the speed of sound is seen/heard to emit a distinct sound. Soldiers marching, tuning radio/TV in electricals

Equation of progressive wave

$$V = \frac{\lambda}{T} = \lambda \nu$$

At a given position in space, the ^{Variation} ~~vibration~~ of the wave amplitude with time is given by

$$y(t) = A \sin \omega t \quad \omega = 2\pi \nu$$

Distance covered is

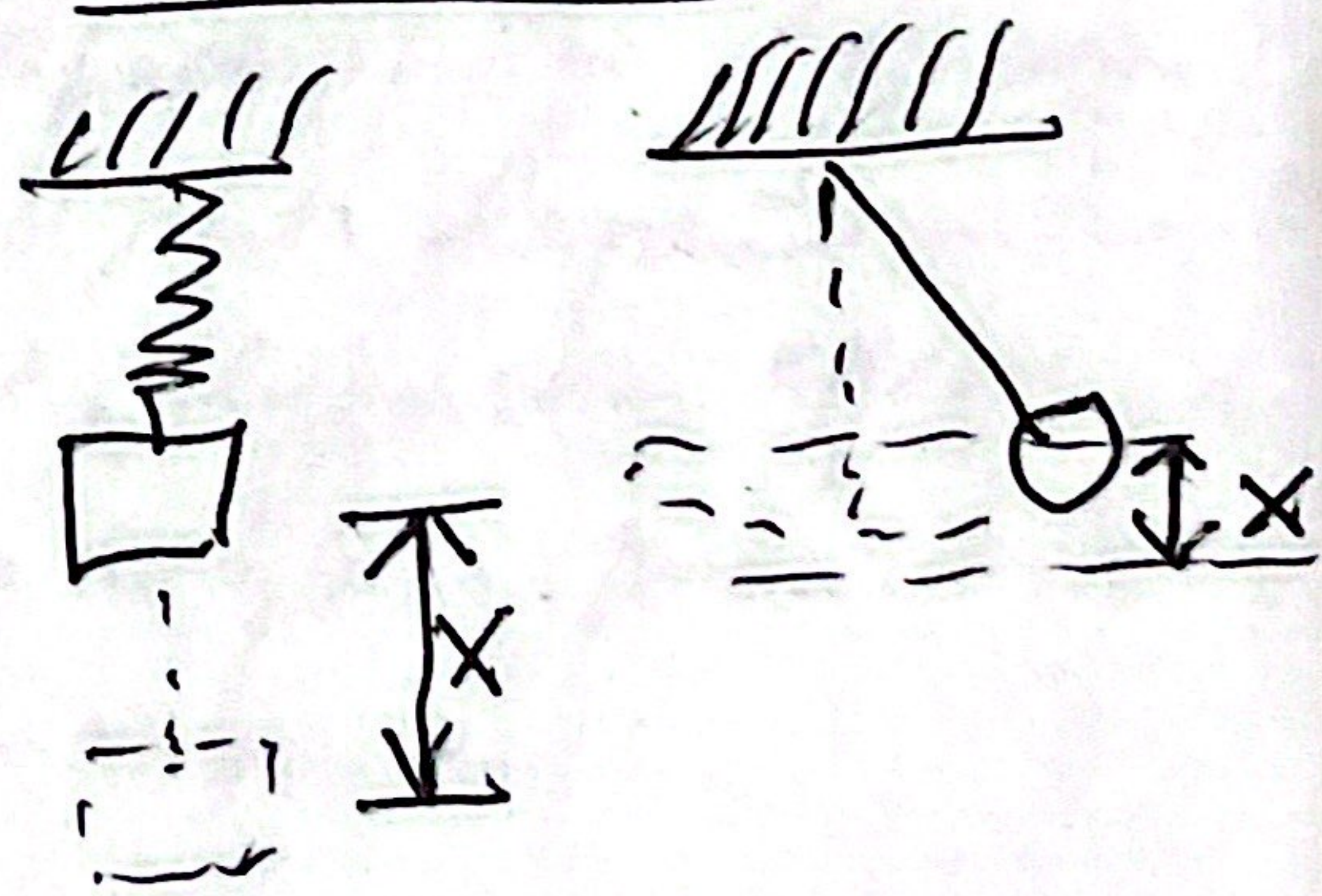
$$x(t) = Vt$$

2nd order ODE

Solution is $x = A \sin \omega t$
like a progressive wave.

$$\omega = \sqrt{\frac{k}{m}}$$

SHM



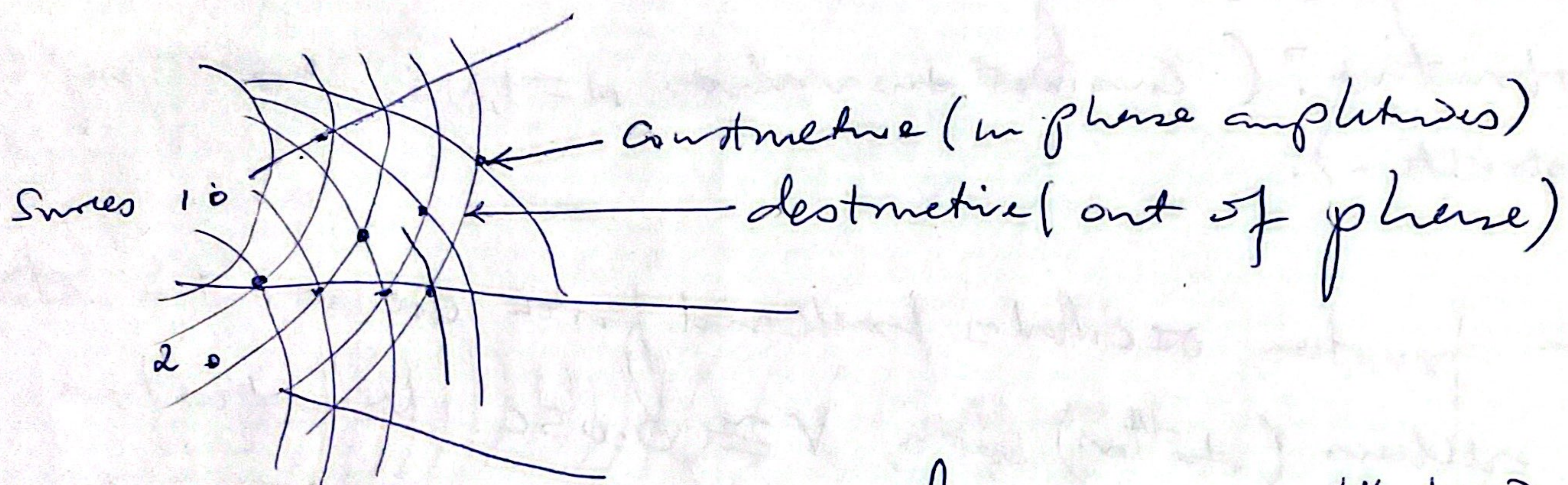
$$F \propto -x$$

$$F = -kx$$

$$m \frac{d^2 x}{dt^2} = -kx$$

Interference

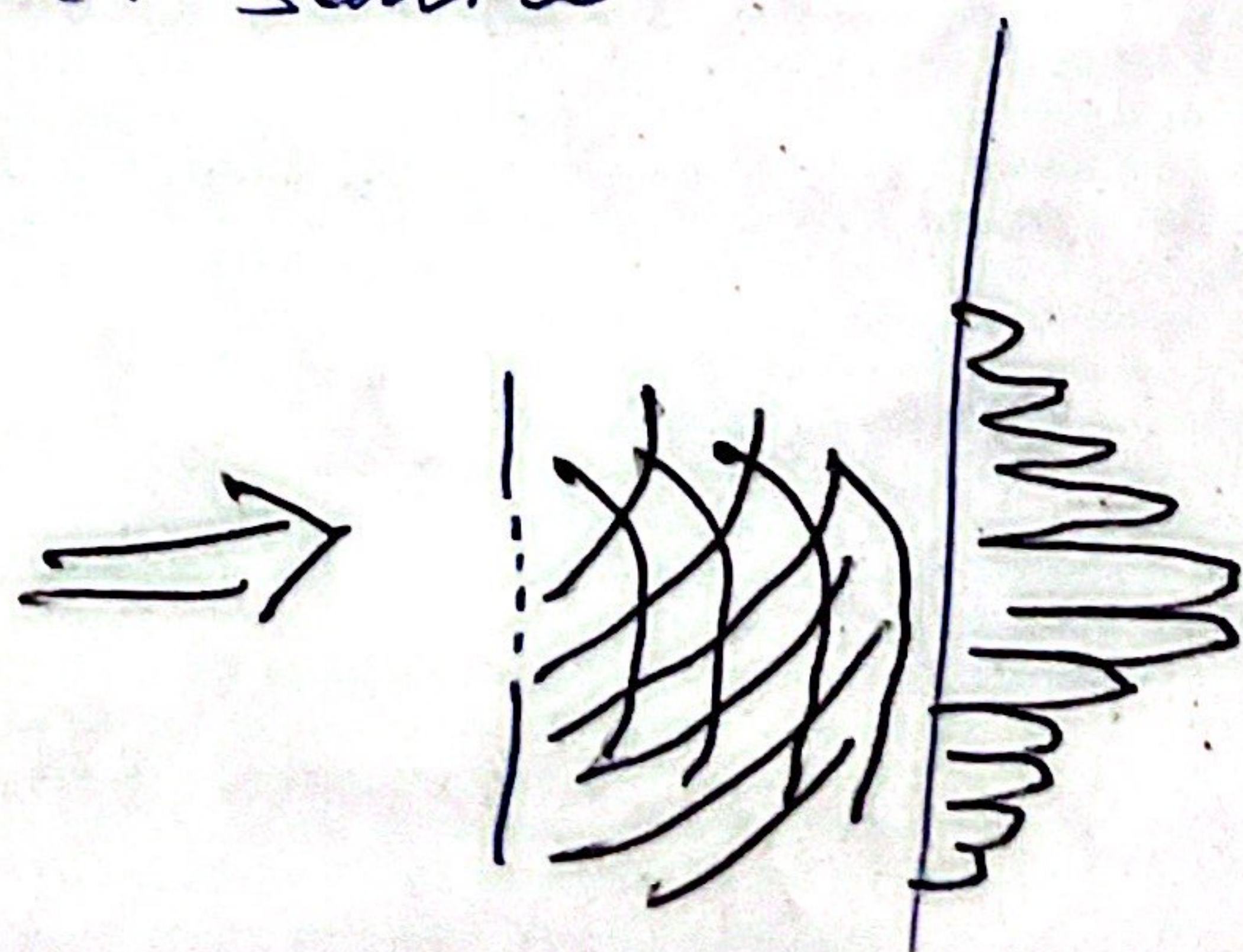
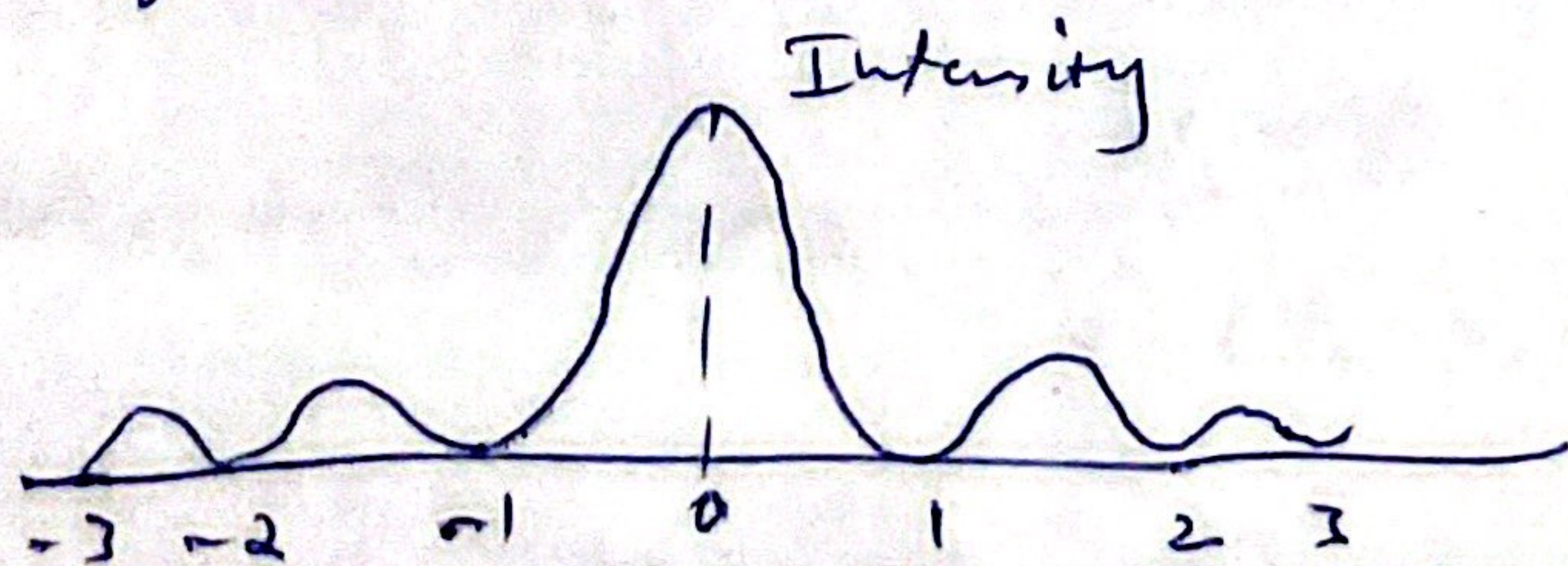
Any wave motion in which the amplitudes of two or more waves combine will exhibit interference. (Constructive).
Wave pulses of opposite sign ~~cancel or destructive interference~~ ^{cancel or (destructive interference)}.



Interference can be in two or 3 dimensions. All kinds of waves interfere (light, sound, molecular, etc). The important property is that, the waves are in phase.

Diffraction

Diffraction causes waves to "bend around" obstacles and to spread out upon passing through apertures. The shape of the diffraction pattern that results when light is passed thru' slits depends on the frequency of the light. Because of this fact, diffraction effects are often used instead of refraction effects in glass prisms to analyse the frequency spectrum of light from a source.

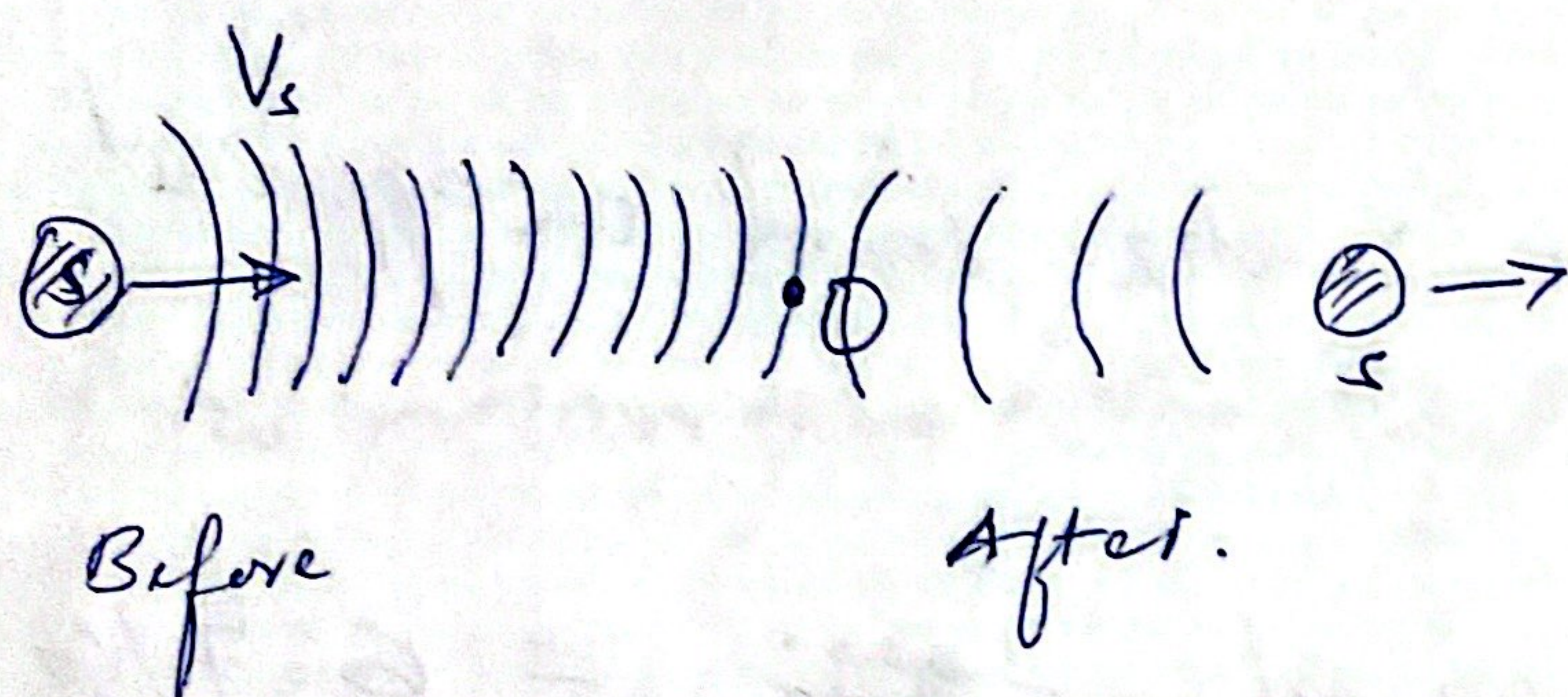


Problems/Exercises.

- 1] As a train passes you hear the frequency of its whistle drop from 1200 Hz to 800 Hz . What is its speed in km/hr ?
- 2] Two sources of waves with $\lambda = 2\text{ m}$ are separated by a distance of 6 m . Along what lines will the interference be constructive? (Consider the values $N = 1, 2, 3, \dots$ use graphical construction).
- 3] The proton oscillates back and forth across the Φ of a nucleus (10^{-14} m) with $v \approx 0.05c$. What is the approximate λ of the emitted radiation? (The frequency of the radiation = to frequency of oscillation) How would you classify this radiation?

Doppler Effect

When a sound source is approaching, the apparent frequency is high, and low when the source passes. The dependence of ν of a wave disturbance on the relative motion of the source and the observer is termed the Doppler effect.



more waves are concentrated when approaching; Therefore the frequency determined by observer is higher

$$\nu_{\text{before}} = \frac{V}{\lambda_0} = V \cdot \frac{V_s}{V - V_s} \quad \text{where} \quad \frac{V_s \lambda_0}{V - V_s} = \Delta \lambda \quad \text{the change in}$$

wavelength.

$$\text{After passing} \quad \nu_{\text{After}} = V \cdot \frac{V_s}{V + V_s} \quad \text{because} \quad \Delta \lambda = \frac{\lambda_0 V_s}{V + V_s}$$

If the observer is moving, then the apparent change in λ is determined by the relative velocity of the source relative to the observer. i.e. $\Delta \lambda = \frac{\lambda_0 V_s}{V - (V_s - V_o)}$ and $\Delta \lambda = \frac{\lambda_0 V_s}{V + (V_s + V_o)}$

for approach and receding cases. If observer approaching then V_o is negative. If in the same direction V_o is positive.

Example.

MT 2 Q 11

An air plane flying at mach 0.5 carries a sound source of 1000 Hz signal. What frequency of sound does the observer hear if he is in the path of the plane?

Sol.

$$V_L = V \cdot \frac{V_s}{V - V_s} = 330 \text{ m/s} \cdot \frac{0.5 \cdot 1000}{330 - 165} = 2000 \text{ Hz}$$

The Listener hears a 2000 Hz sound on approach.

$$V_L = V \cdot \frac{V_s}{V + V_s} = 330 \text{ m/s} \cdot \frac{1000}{330 + 165} = 667 \text{ Hz}$$

The Listener hears a 667 Hz sound on recede.

* Suppose the Listener was moving in the direction of the plane at 10 m/s. And in the opposite direction at the same speed.